Hope Enterprise University College	
Faculty of Informatics	
Department of Computer Science	
Course Outline	
Course Title:	**Multimedia Systems**
Course Code:	**CS413**
Credit Hours:	**3**
Contact Hours:	**4**
Lab Hours:	**2**
Prerequisite:	**Data Structures and Algorithm Analysis**
Course Category:	**Elective**
Year:	**4**
Semester:	**1**
Instructor	**Gulelat Yohannes**

Course Description

Multimedia data has become an indispensable part of our daily life. It is also one of the critical applications in broad areas of use. In this course students will be introduced to the principles and current technologies of multimedia systems. The course includes the topics introduction to multimedia and multimedia systems, multimedia data representation, multimedia applications, multimedia tools, hands on practice on multimedia system creation using tools, multimedia standards, communication requirements of multimedia data and multimedia information retrieval.

Course Objectives

At the end of the course students should be able to:

- Understand principles and current technologies of multimedia systems,
- Comprehend applications of multimedia systems in day-to-day life
- Design and develop electrifying multimedia rich contents for various application domains like web sites and databases

Course Contents	Weeks

Chapter 1: Introduction to Multimedia (8 hours/3 weeks)	
1.1 What is Multimedia? 1.2 Multimedia Usage or Application 1.3 Advantage and Disadvantage of using Multimedia 1.4 Multimedia Categories 1.5 Multimedia System 1.5.1 Characteristics of Multimedia systems 1.5.2 Components of Multimedia systems 1.5.3 Major challenges and Desirable Properties of MM Systems	**1-3**
Chapter 2: Issues in Multimedia Authoring(10 hours/ 5 weeks)	
2.1. Introduction to Multimedia Document Model 2.2. What is Multimedia Authoring? 2.2.1. MM Authoring Paradigms 2.3. Authoring Dimensions 2.4.Multimedia Authoring Tools 2.4.1. Types of Authoring tools 2.5.Multimedia Production Process 2.5.1. Preproduction 2.5.2. Production 2.5.3. Postproduction	**4 - 8**
Mid Exam Week	**9**
Chapter 3: Multimedia Data Representation(8 hours/ 4 weeks)	
3.1. Text in Multimedia 3.2. Sound in Multimedia 3.3. Images in Multimedia 3.4. Video and Animation in Multimedia	**10 - 13**
Chapter 4: Multimedia Data Compression (6hrs / 3 Weeks)	
4.1.Introduction to Multimedia Data Compression 4.2. Types of Multimedia Data compression Techniques 4.2.1. Lossless Compression Technique 4.2.2. Lossy Compression Technique	**14 - 16**

Chapter 5 : Adobe Flash or Adobe Animate – Lab Session (24 hrs /12 weeks)	
5.1. Introduction to Adobe Animate or Flash CS4/6 5.2. Basic Screen elements and Customization 5.3. Using Tools panel to add and/ or modify objects 5.4. Using Timeline and Creating different types of frames and types of layers 5.5. Creating reusable objects or symbols 5.5.1. Creating Graphic symbol 5.5.2. Creating Movie Clip Symbol 5.5.3. Creating Button symbol 5.6. Animation 5.6.1. The 12 Principles of Animation 5.6.2. Types of Animation in Animate 5.6.3. Creating Animation 5.7. Adding Interactivity	**4 - 16**

Teaching and Learning Method:

Lecture, Lab Exercises, Individual and group assignments and mini project work

Assessment /Evaluation Scheme

Component	Weight	Coverage	Assessment Time		
Mid Exam	**20%**	**Chapter 1 - 2**	**Week 9**		
Individual Assignment	15%	-	Week 3 and Week 6		
Project Work	15%	Chapter 1 - 5	Week 7, Week 10 and Week 14		
Final Exam	**50%**	**All Chapters**			
Total	100%				

Text Book

Tay Vaughan. 2006. Multimedia: Making it Work, 7 th ed. ISBN-10 0072264517.

Reference Books

Suzanne, Jennifer, et el., Multimedia Basics, 2004, Thomson Course Tech.

Ralf Steinmetz and Klara Nahrstedt, Multimedia Fundamentals: Media Coding and Content Processing; Prentice Hall